Predictive Statistics for Business Decision Making Report

Contents

1. **Problem 1 .....3**
   1.1 What is the probability that a randomly chosen player would suffer an injury?
   1.2 What is the probability that a player is a forward or a winger?
   1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?
   1.4 What is the probability that a randomly chosen injured player is a striker?
2. **Problem 2 .....4**
   2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?
   2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?
   2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?
   2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?
3. **About the Data and Its Dictionary 7**
4. **Univariate Analysis**
5. **Bivariate Analysis**
6. **Problem 3 ..... 19**
   3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?
   3.2 Is the mean hardness of the polished and unpolished stones the same?
7. **Problem 4..... 20**
   4.1 How does the hardness of implants vary depending on dentists?
   4.2 How does the hardness of implants vary depending on methods?
   4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?
   4.4 How does the hardness of implants vary depending on dentists and methods together?

Problem 1

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Striker	Forward	Attacking Midfielder	Winger	**Total**
Players Injured	45	56	24	20	**145**
Players Not Injured	32	38	11	9	**90**
Total	**77**	**94**	**35**	**29**	**235**

1.1 What is the probability that a randomly chosen player would suffer an injury?

P(Injury) = Total Players / Players Injured = 145/235 = 0.617 = 62%
The probability that a randomly chosen player would suffer an injury is 62 %

1.2 What is the probability that a player is a forward or a winger?

P(forward or winger) = Total Forwards + Total Wingers / Total Players = 94+29/ 235 = 0.523 = 52%
The probability that a player is a forward or a winger is 52%

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

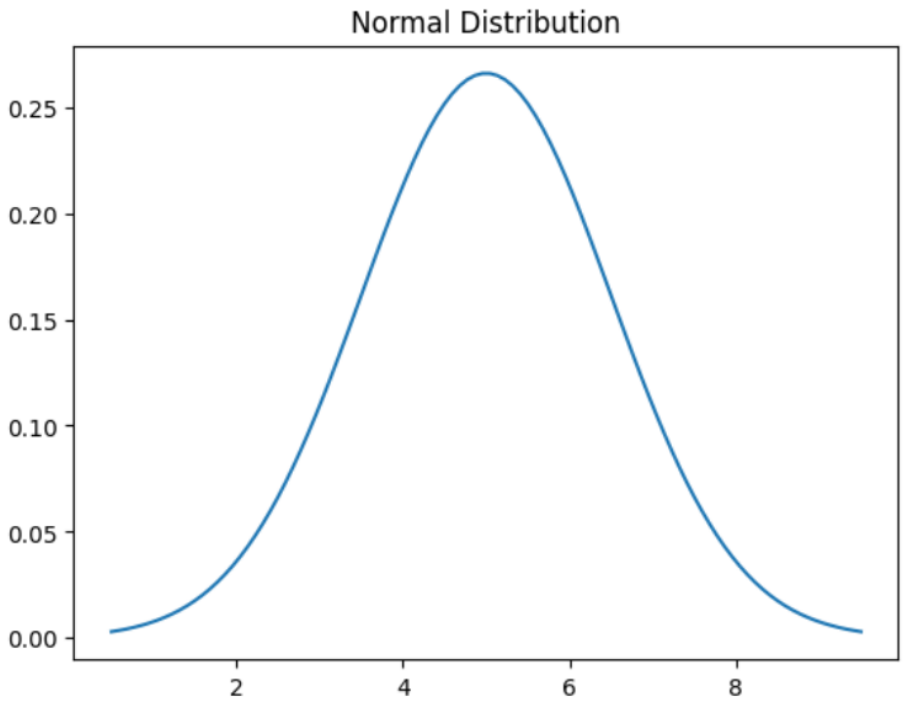
P(Striker and Injury) = Striker Injured/Total Players = 45/235 = 0.191 = 19%
The probability that a randomly chosen player plays in a striker position and has a foot injury is 19%

1.4 What is the probability that a randomly chosen injured player is a striker?

P(Striker | Injury) = Striker Injured/Players Injured = 45/145 = 0.310 = 31%
The probability that a randomly chosen injured player is a striker is 31%

Problem 2

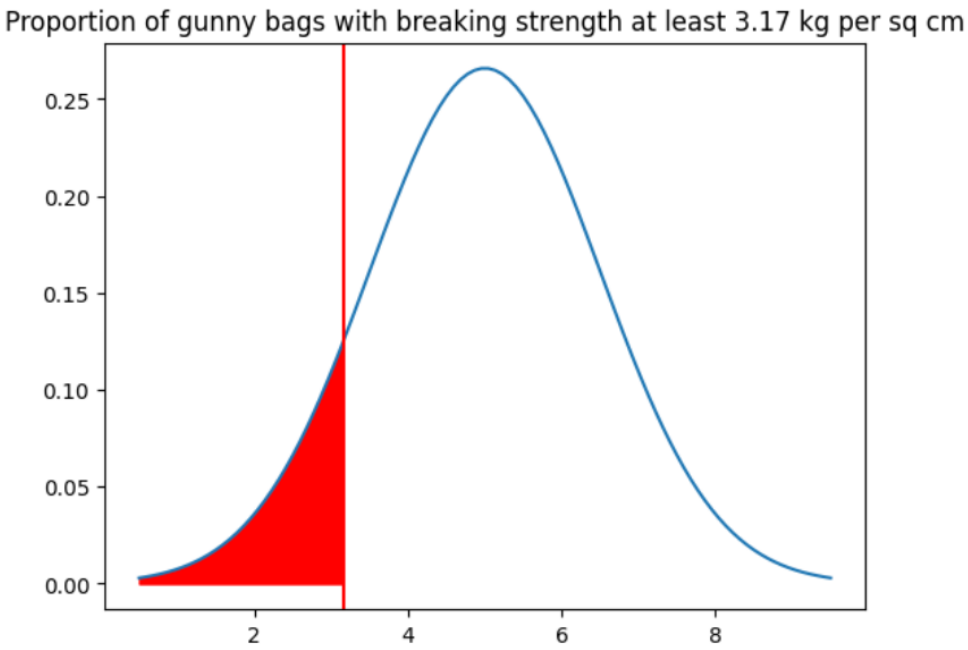
The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain.



2.1 What proportion of the gunny bags have a breaking strength of

less than 3.17 kg per sq cm?

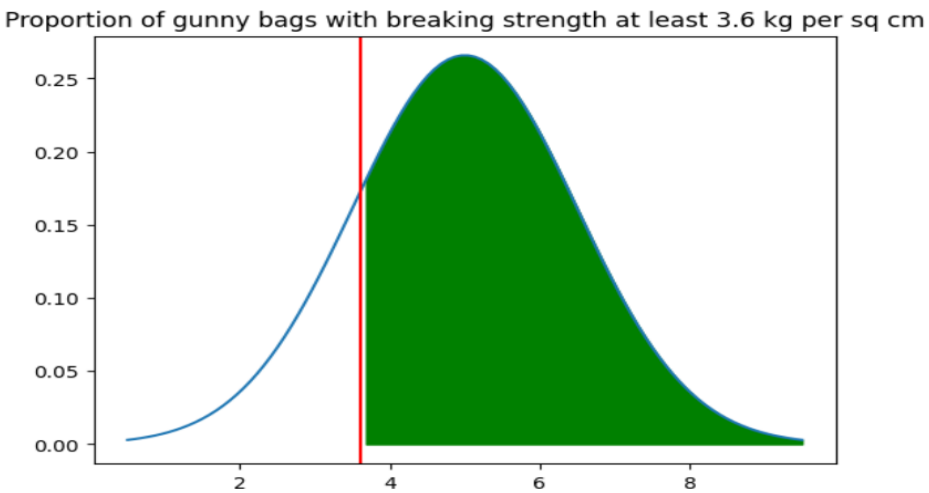
Proportion of gunny bags with breaking strength at least 3.17 kg per sq cm: 0.11123



This plot shows the normal distribution and highlights the area under the curve for breaking strengths less than 3.17 kg per sq cm.

2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?

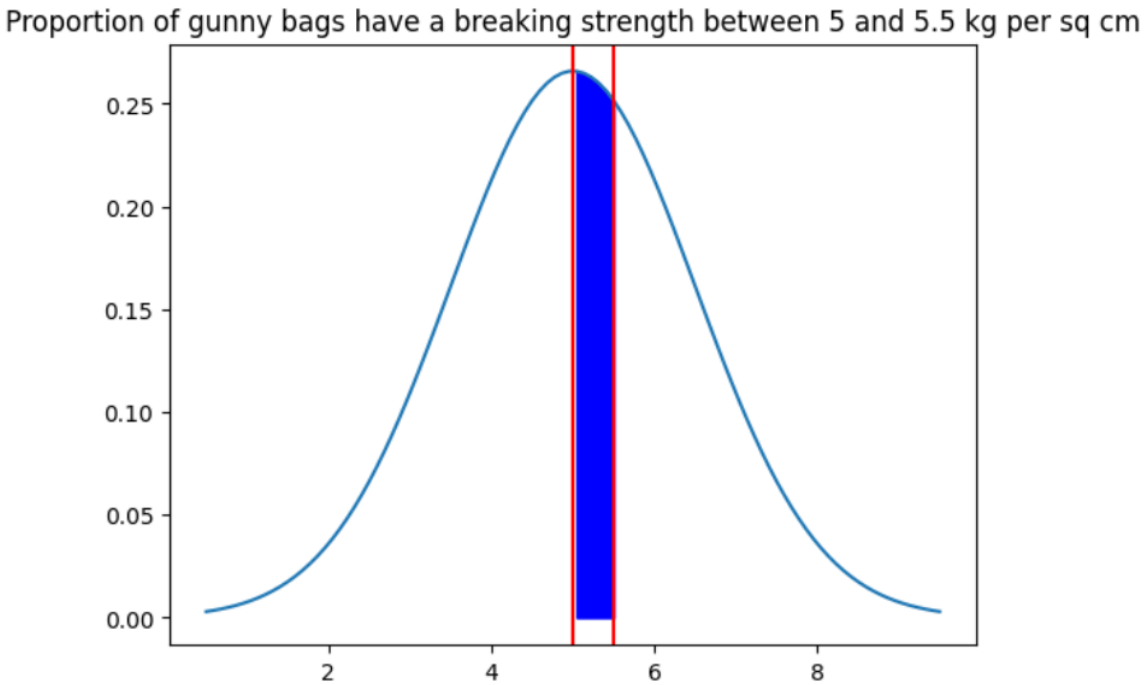
Proportion of gunny bags with breaking strength at least 3.6 kg per sq cm: 0.8247



This plot shows the normal distribution and highlights the area under the curve for breaking strengths of at least 3.6 kg per sq cm.

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?

Proportion of gunny bags have a breaking strength between 5 and 5.5 kg per sq cm: 0.1306

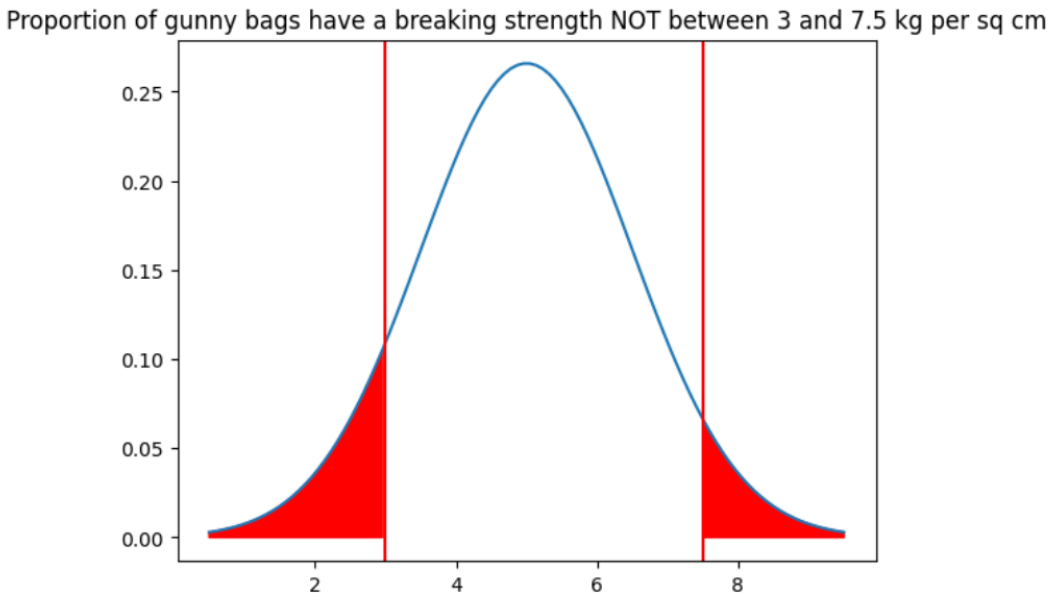


This plot shows the normal distribution and highlights the area under the curve for breaking strengths between 5 and 5.5 kg per sq cm.

2.4 What proportion of the gunny bags have a breaking strength

NOT between 3 and 7.5 kg per sq cm.?

Proportion of gunny bags have a breaking strength NOT between 3 a nd 7.5 kg per sq cm: 0.139



This plot shows the normal distribution and highlights the area under the curve for breaking strengths not between 3 and 7.5 kg per sq cm.

About Data and Dictionary

There are two Datasets:

1. Zingaro company
2. Dental Hardness

1. Zingaro company

Data Overview:
The dataset provides comprehensive details about Unpolished and the Treated & Polished information.

Number of rows and columns:

```
The number of rows in data: 75
The number of columns in data: 2
```

First Five Rows:

	Unpolished	Treated and Polished
0	164.481713	133.209393
1	154.307045	138.482771
2	129.861048	159.665201
3	159.096184	145.663528
4	135.256748	136.789227

DataFrame Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 75 entries, 0 to 74
Data columns (total 2 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Unpolished            75 non-null     float64
 1   Treated and Polished  75 non-null     float64
```

Checking Null values:

```
Unpolished              0
Treated and Polished    0
dtype: int64
```

Checking Duplicate values:

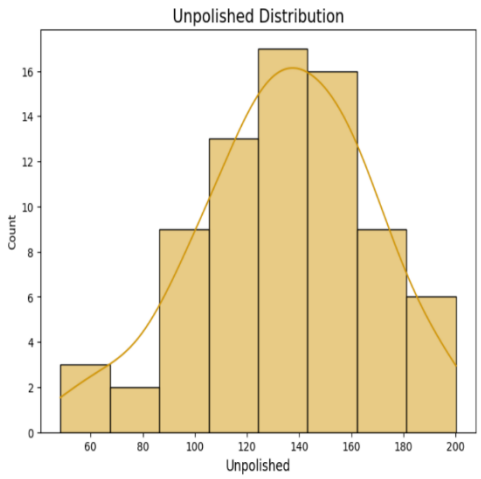
```
0
```

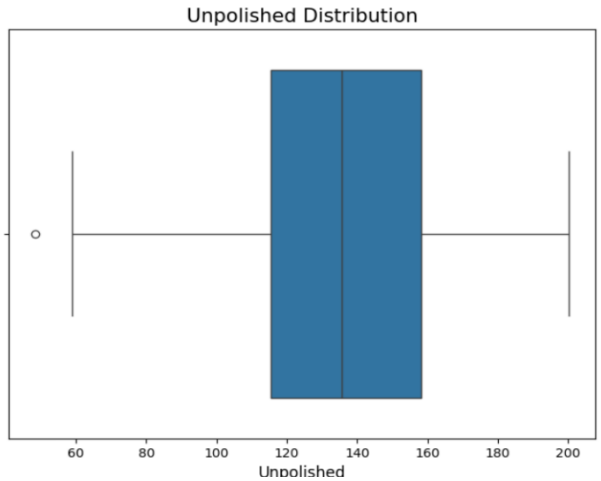
Statistical Information:

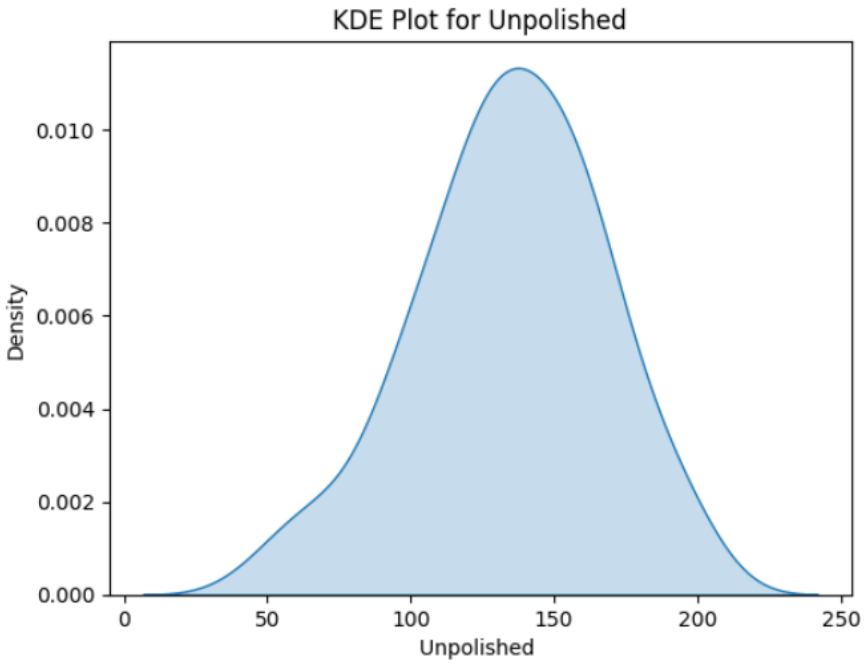
	count	mean	std	min	25%	50%	75%	max
Unpolished	75.0	134.110527	33.041804	48.406838	115.329753	135.597121	158.215098	200.161313
Treated and Polished	75.0	147.788117	15.587355	107.524167	138.268300	145.721322	157.373318	192.272856

Univariate Analysis for Zingaro Company

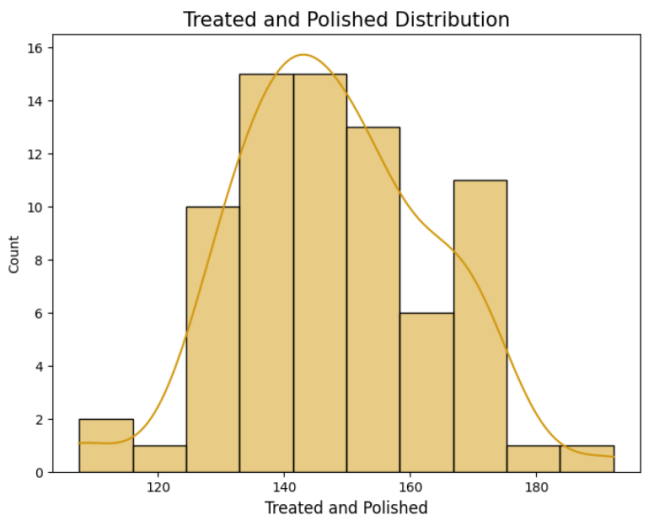
Unpolished Distribution:

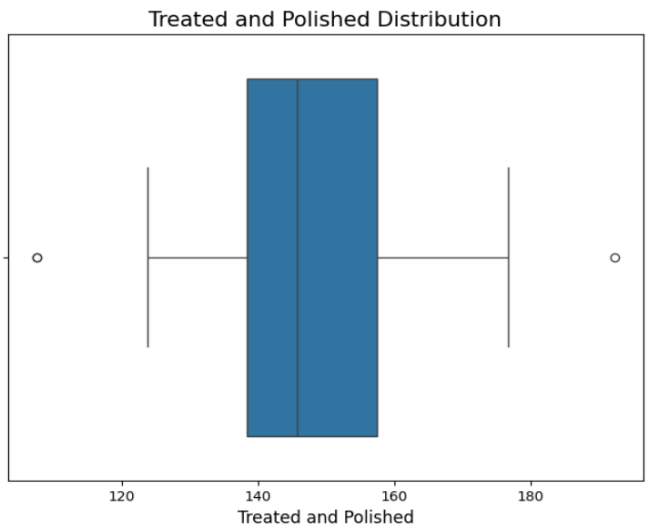


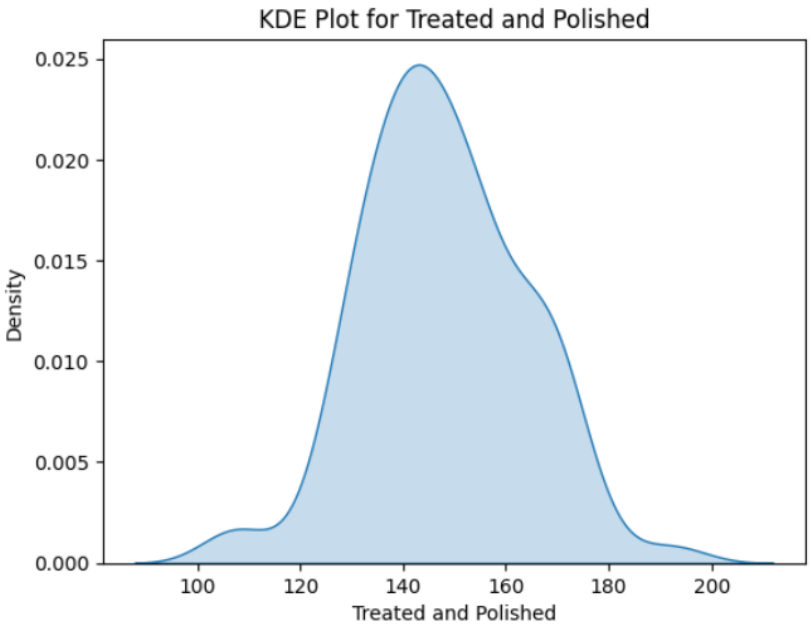




Treated and Polished Distribution:

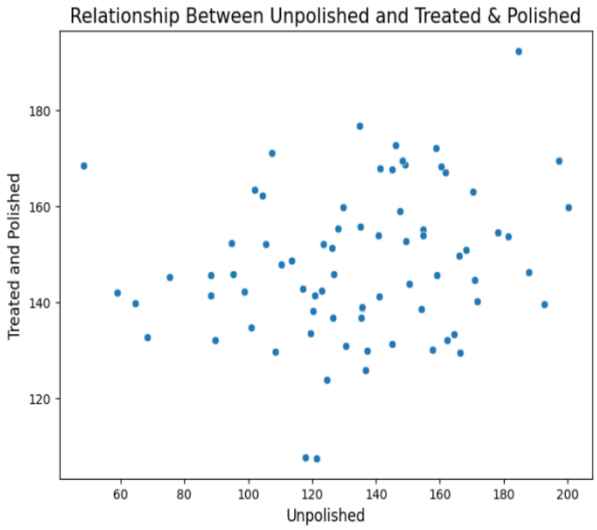


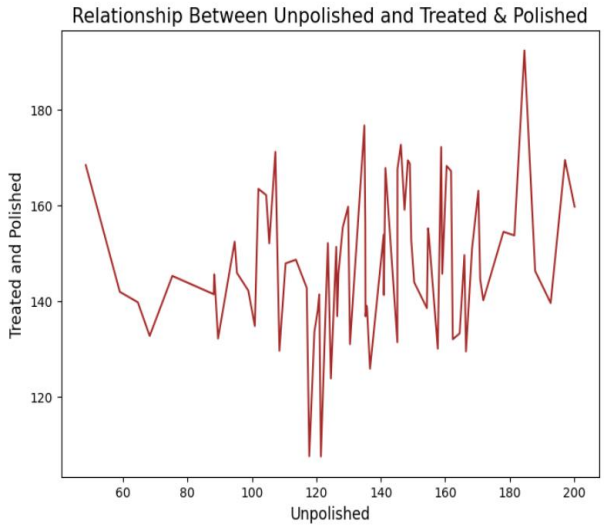




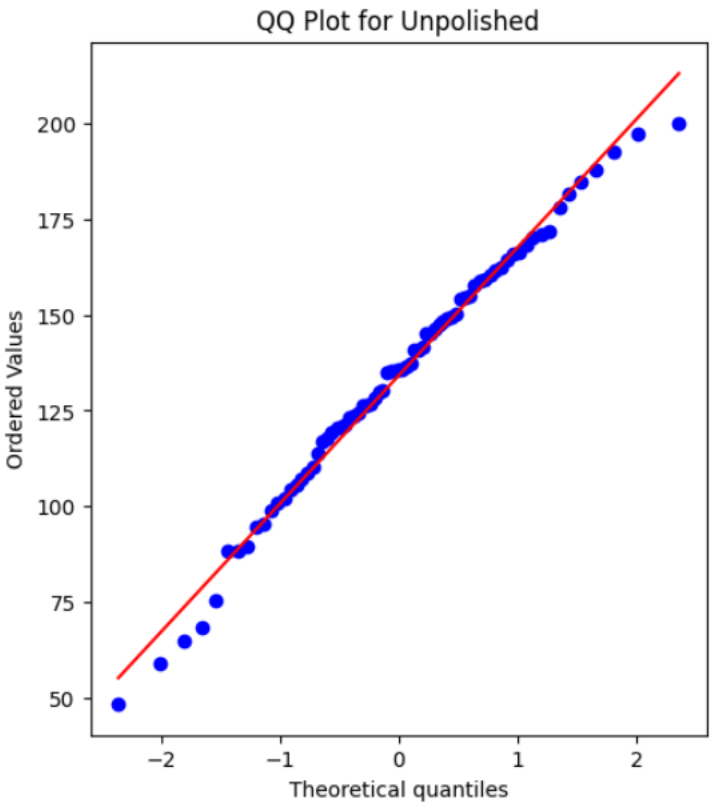
Bivariate Analysis for Zingaro Company

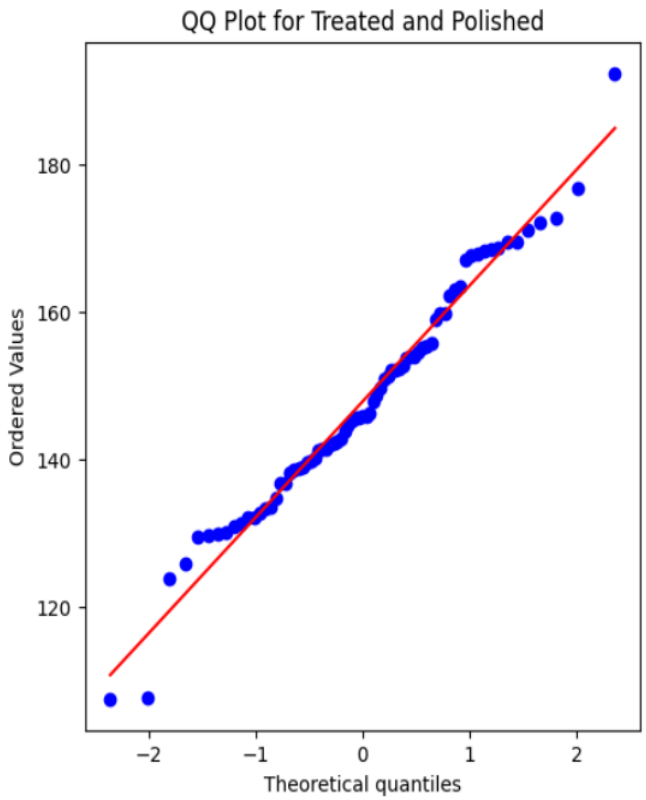
Relationship Between Unpolished and Treated & Polished:





QQ Plot for Unpolished and Treated & Polished:





2. Dental Hardness

Data Overview:
The dataset provides comprehensive details about:

1. **Dentist**: Identifier for the dentist
2. **Method**: Method used
3. **Alloy**: Type of alloy used
4. **Temp**: Temperature in degrees
5. **Response**: Hardness response

Number of rows and columns:

```
The number of rows in data: 90
The number of columns in data: 5
```

First Five Rows:

	Dentist	Method	Alloy	Temp	Response
0	1	1	1	1500	813
1	1	1	1	1600	792
2	1	1	1	1700	792
3	1	1	2	1500	907
4	1	1	2	1600	792

Checking Null values:

```
Dentist     0
Method      0
Alloy       0
Temp        0
Response    0
dtype: int64
```

Data frame Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Dentist   90 non-null     int64
 1   Method    90 non-null     int64
 2   Alloy     90 non-null     int64
 3   Temp      90 non-null     int64
 4   Response  90 non-null     int64
```

Checking Duplicate values:

```
0
```

Dentist Value Counts:

```
Dentist
1    18
2    18
3    18
4    18
5    18
Name: count, dtype: int64
```

Method Value Counts:

```
Method
1    30
2    30
3    30
Name: count, dtype: int64
```

Alloy Value Counts:

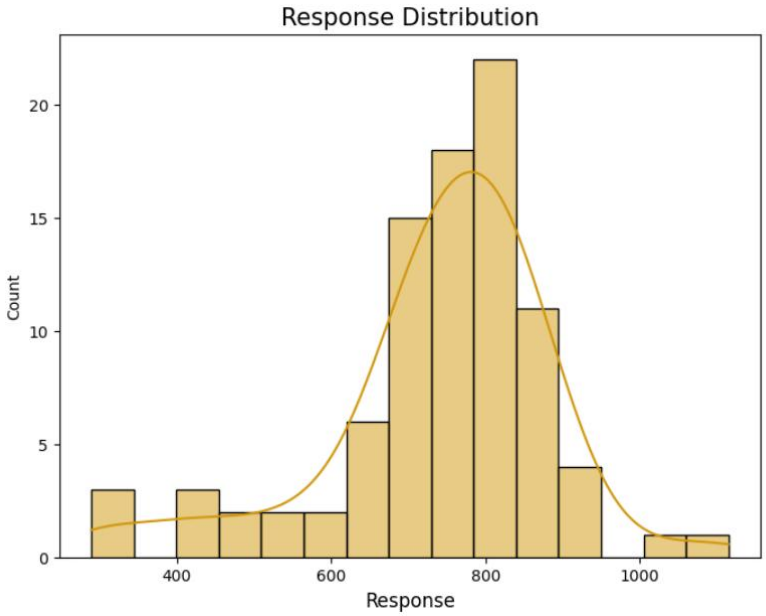
```
Alloy
1    45
2    45
Name: count, dtype: int64
```

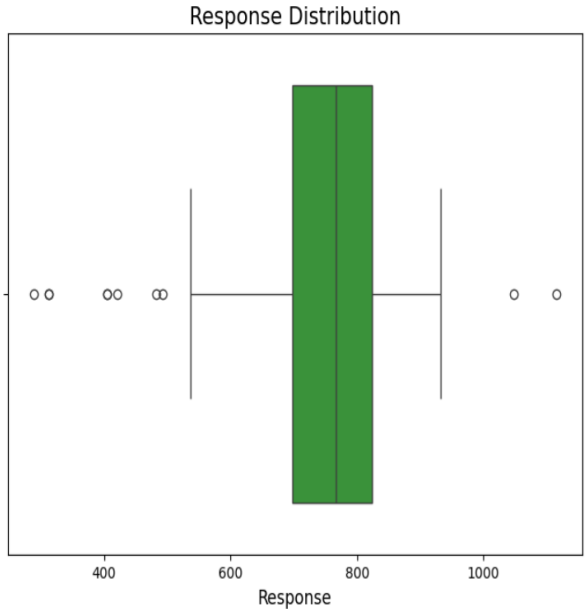
Statistical Information:

	count	mean	std	min	25%	50%	75%	max
Dentist	90.0	3.000000	1.422136	1.0	2.0	3.0	4.0	5.0
Method	90.0	2.000000	0.821071	1.0	1.0	2.0	3.0	3.0
Alloy	90.0	1.500000	0.502801	1.0	1.0	1.5	2.0	2.0
Temp	90.0	1600.000000	82.107083	1500.0	1500.0	1600.0	1700.0	1700.0
Response	90.0	741.777778	145.767845	289.0	698.0	767.0	824.0	1115.0

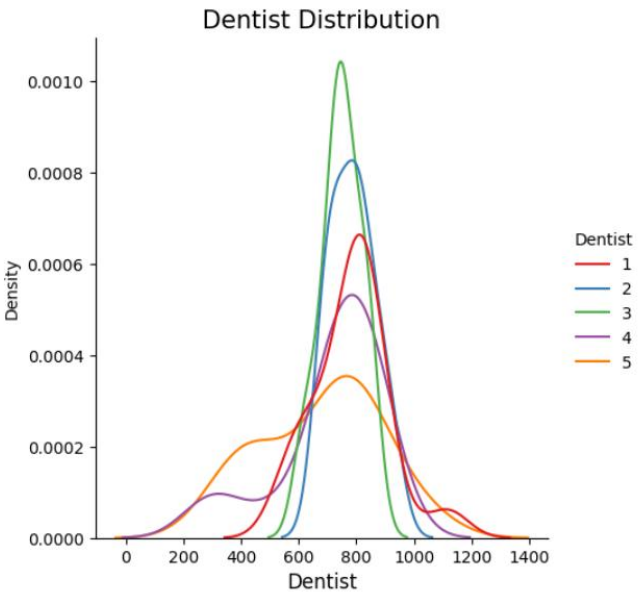
Univariate Analysis for Dental Hardness

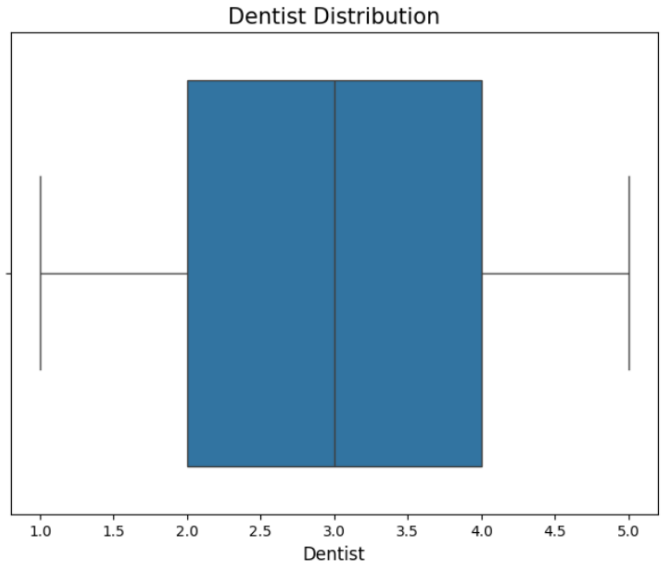
Response Distribution:



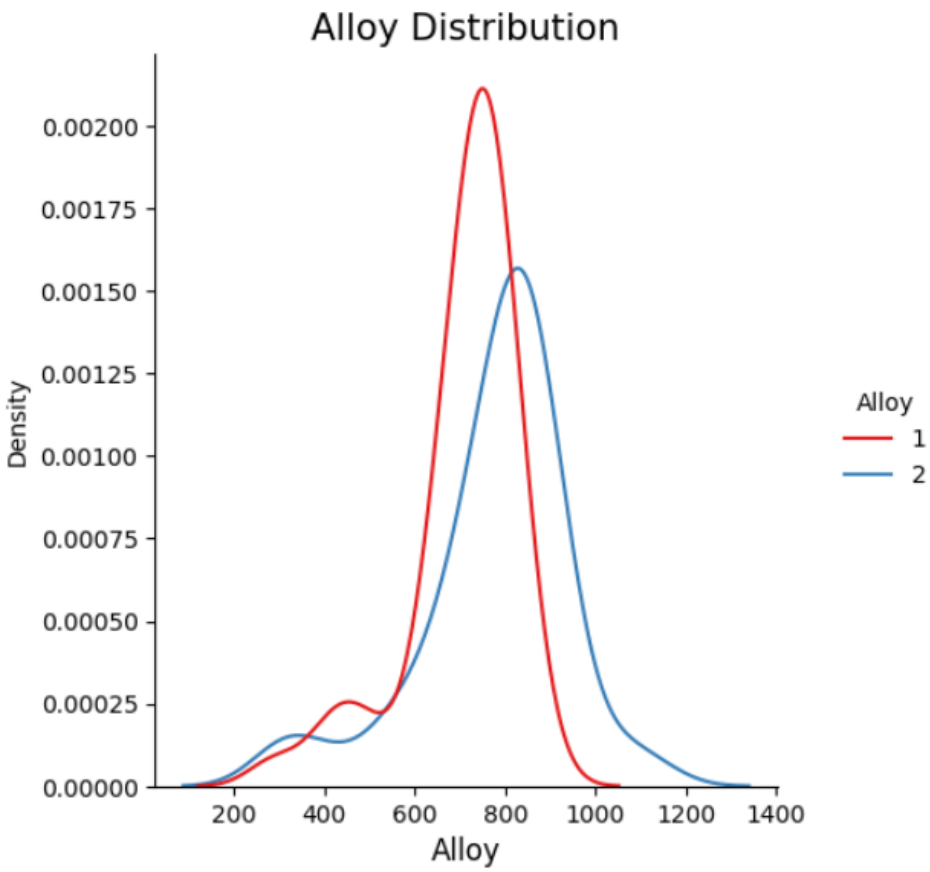


Dentist Distribution:

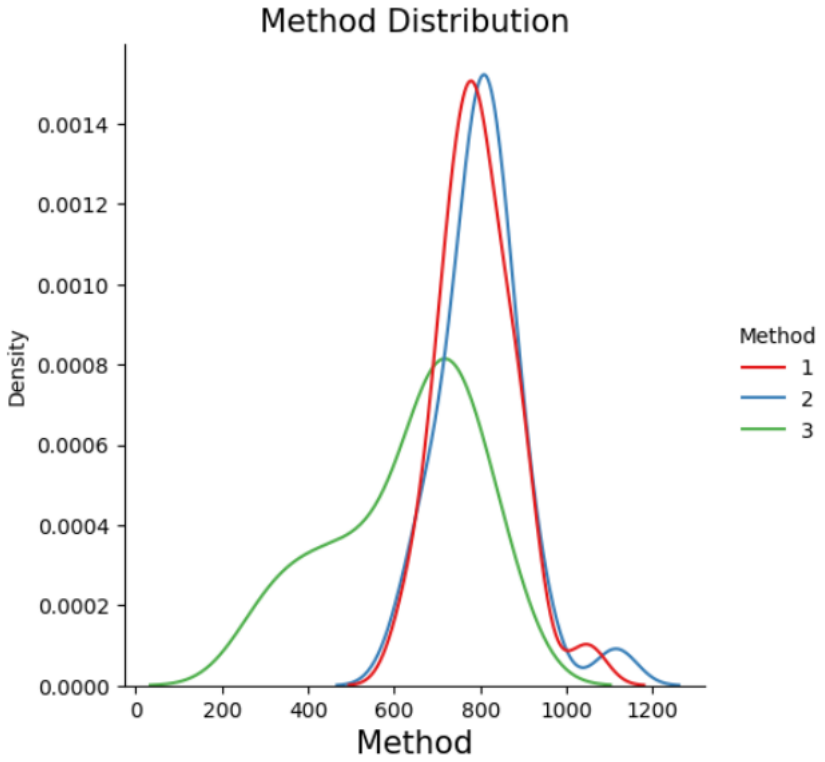




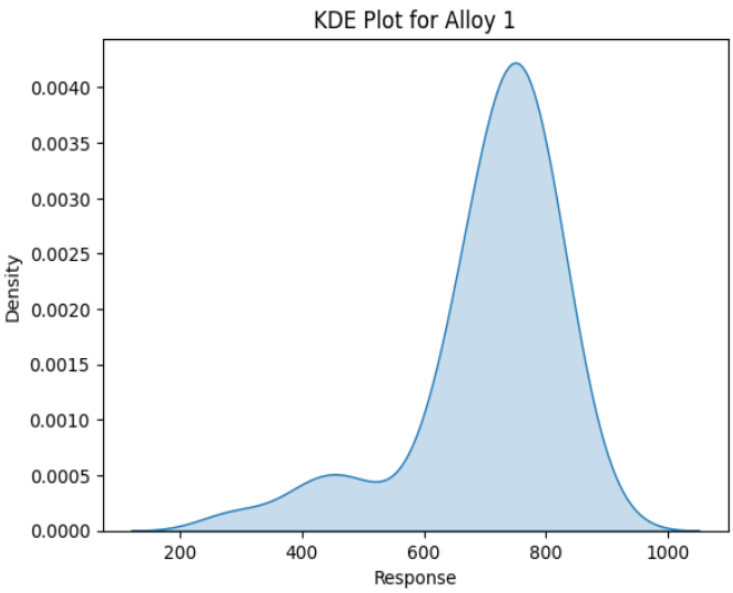
Alloy Distribution:

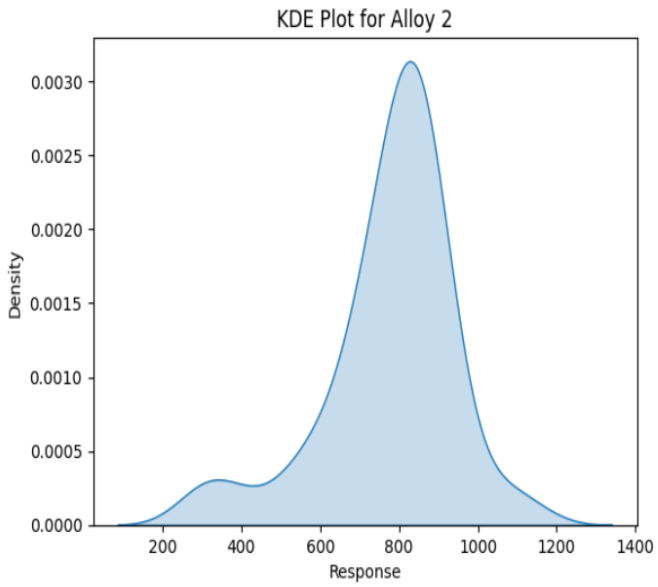


Method Distribution:

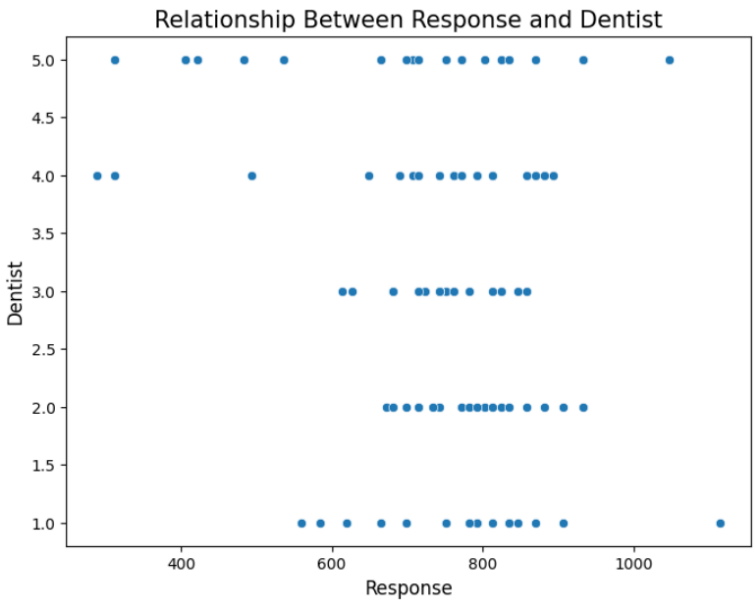


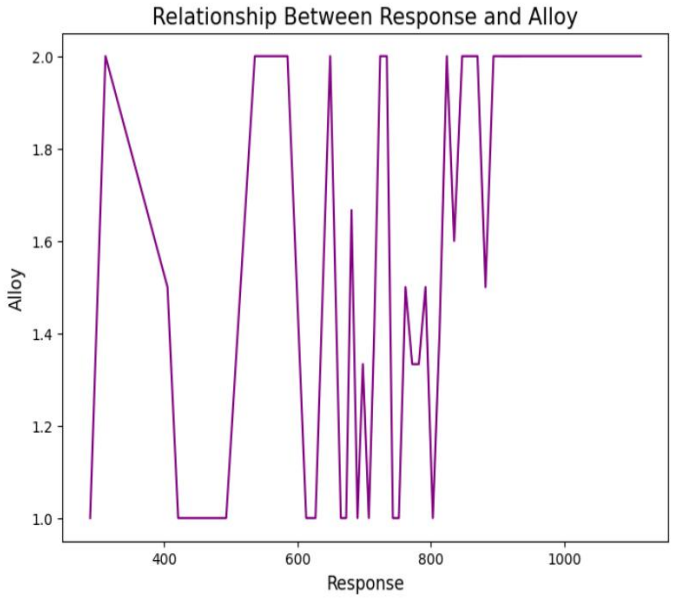
KDE plot for Alloy 1 and Alloy 2:

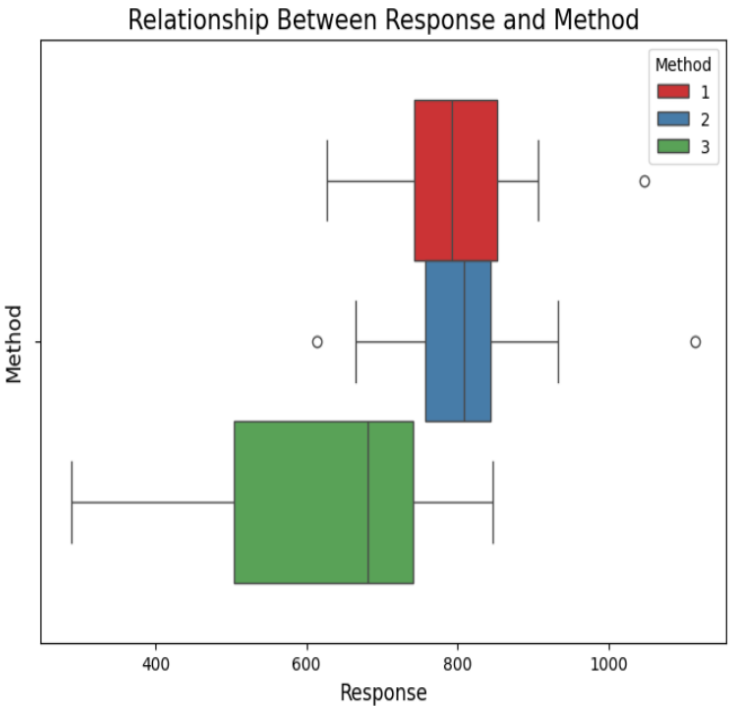


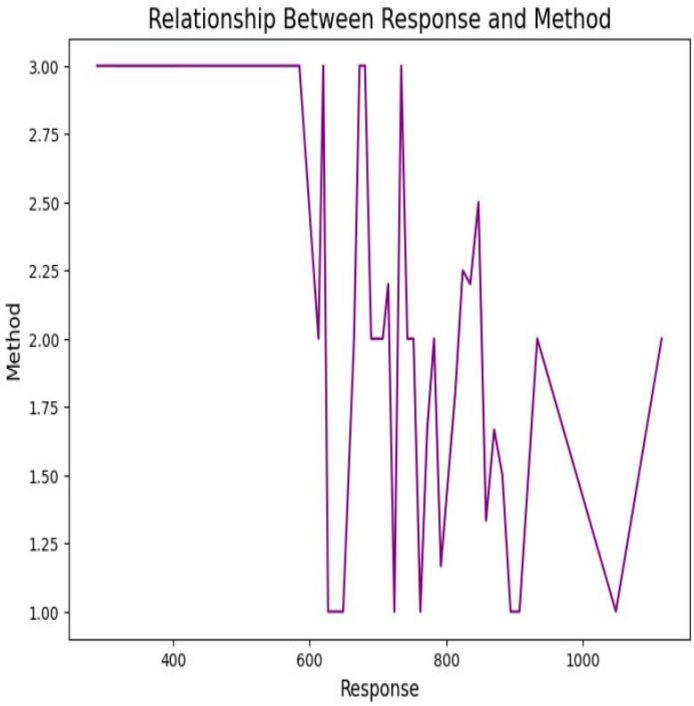


Bivariate Analysis for Dental Hardness

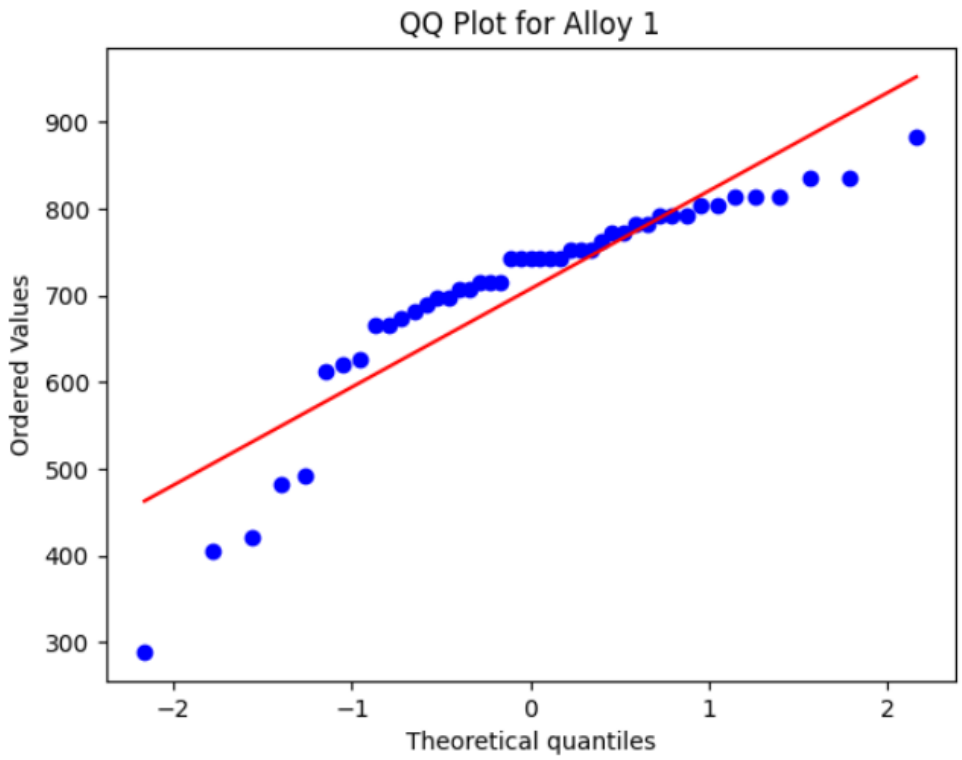




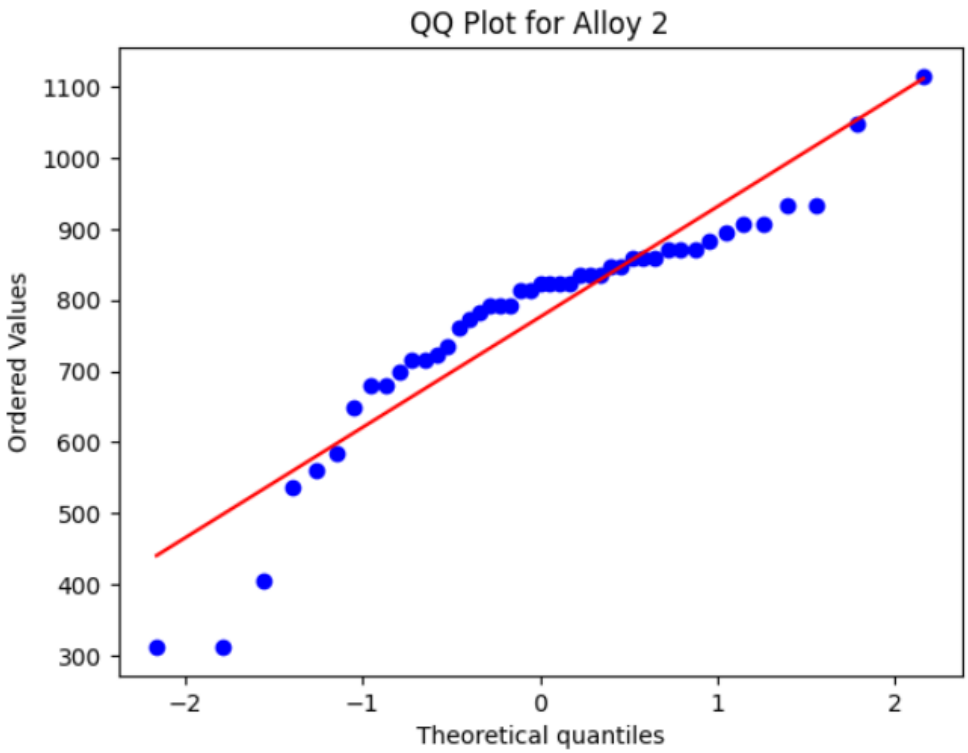




QQ-plot for Alloy 1:



QQ-plot for Alloy 2:



Problem 3

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level).

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

Test Used: One-sample t-test

Hypothesis:

- **Null Hypothesis (H0):** The mean hardness of unpolished stones is equal to 150.
- **Alternative Hypothesis (H1):** The mean hardness of unpolished stones is less than 150.

Test Results:

- **T-statistic**: -4.16463
- **P-value**: 4.171286997419652e-05

Conclusion:

Since the p-value is less than the significance level (0.05), we **reject the null hypothesis**. This means that Zingaro is justified in thinking that the unpolished stones may not be suitable for printing.

3.2 Is the mean hardness of the polished and unpolished stones the same?

Test Used: Independent two-sample t-test

Hypothesis:

- **Null Hypothesis (H0):** The mean hardness of polished and unpolished stones is the same.
- **Alternative Hypothesis (H1):** The mean hardness of polished and unpolished stones is different.

Test Results:

- **T-statistic**: -3.24223
- **P-value**: 0.00147

Conclusion:

Since the p-value (0.0015) is less than the significance level (0.05), we **reject the null hypothesis**. This means that the mean hardness of polished and unpolished stones is not the same.

Problem 4

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favor one method above another and may work better in his/her favorite method. The response is the variable of interest.

4.1 How does the hardness of implants vary depending on dentists?

Hypothesis:

- **Null Hypothesis (H0):** The mean hardness of implants is the same for all dentists.
- **Alternative Hypothesis (H1):** The mean hardness of implants is different for at least one dentist.

Analysis for Alloy 1:

Shapiro Test for Normality:

Dentist	P-value
1	0.3254
2	0.8415
3	0.1295
4	0.0533
5	0.0812

Conclusion:

Since p-values are greater than 0.05, indicating that the data for each dentist is normally distributed. Therefore, we can assume normal distribution for Alloy type 1.

Levene's Test for Homogeneity of Variances:

- **Statistic**: 1.3847
- **P-value**: 0.2565

Conclusion:

Since p-values are greater than 0.05, all p-values are greater than 0.05, indicating that all population variances are equal. Therefore, we can assume equal variance for Alloy type 1.

One-way ANOVA test:

Test Results:

- **Statistic**: 1.9771
- **P-value**: 0.1165

Conclusion:

Since the p-value is greater than the level of significance (5%), we **fail to reject the null hypothesis**. Hence, we do not have enough statistical evidence to say that the mean hardness of implants is different for at least one dentist.

Analysis for Alloy 2:

Shapiro Test for Normality:

Dentist	P-value
1	0.2759
2	0.5735
3	0.5213
4	0.0073
5	0.3386

Conclusion:

Since all p-values are greater than 0.05 with the exception of Dentist 4, which rejects the null hypothesis, indicating that the data for Dentist 4 is not normally distributed, whereas the other Dentist p-values are greater, indicating that the data is normally distributed.

Levene's Test for Homogeneity of Variances:

- **Statistic**: 1.4456
- **P-value**: 0.2368

Conclusion:

Since p-values are greater than 0.05, all p-values are greater than 0.05, indicating that all population variances are equal. Therefore, we can assume equal variance for Alloy type 2.

One-way ANOVA test:

Test Results:

- **Statistic**: 0.5248
- **P-value**: 0.7180

Conclusion:

Since the p-value is greater than the level of significance (5%), we **fail to reject the null hypothesis**. Hence, we do not have enough statistical evidence to say that the mean hardness of implants is different for at least one dentist.

4.2 How does the hardness of implants vary depending on methods?

Hypothesis:

- **Null Hypothesis (H0):** The mean hardness of implants is the same for all methods.
- **Alternative Hypothesis (H1):** The mean hardness of implants is different for at least one method.

Analysis for Alloy 1:

Shapiro Test for Normality:

Method	P-value
1	0.1819
2	0.9030
3	0.1425

Conclusion:
Since p-values are greater than 0.05, indicating that the data for each

Method is normally distributed. Therefore, we can assume normal distribution for Alloy type 1.

Levene's Test for Homogeneity of Variances:

- **Statistic**: 6.5214
- **P-value**: 0.0034

Conclusion:
Since p-value is below 0.05, we reject the null hypothesis of equal variance. This means the implant hardness variances across methods are significantly different for Alloy 1.

One-way ANOVA test:

Test Results:

- **Statistic**: 6.2633
- **P-value**: 0.00416

Conclusion:

Since the p-value is less than the level of significance (5%), we **Reject the null hypothesis**. Hence, we have enough statistical evidence to say that the mean hardness of implants is different for at least one Method.

Analysis for Alloy 2:

Shapiro Test for Normality:

Method	P-value
1	0.7582
2	0.00105
3	0.1025

Conclusion:

Since all p-values are greater than 0.05 with the exception of Method 2, which rejects the null hypothesis, indicating that the data for Method 2 is not normally distributed, whereas the other Method p-values are greater, indicating that the data is normally distributed.

Levene's Test for Homogeneity of Variances:

- **Statistic**: 3.3497
- **P-value**: 0.0446

Conclusion:

Since p-value is below 0.05, we reject the null hypothesis of equal variance. This means the implant hardness variances across methods are significantly different for Alloy 2.

One-way ANOVA test:

Test Results:

- **Statistic**: 6.2633
- **P-value**: 5.415871051443187e-06

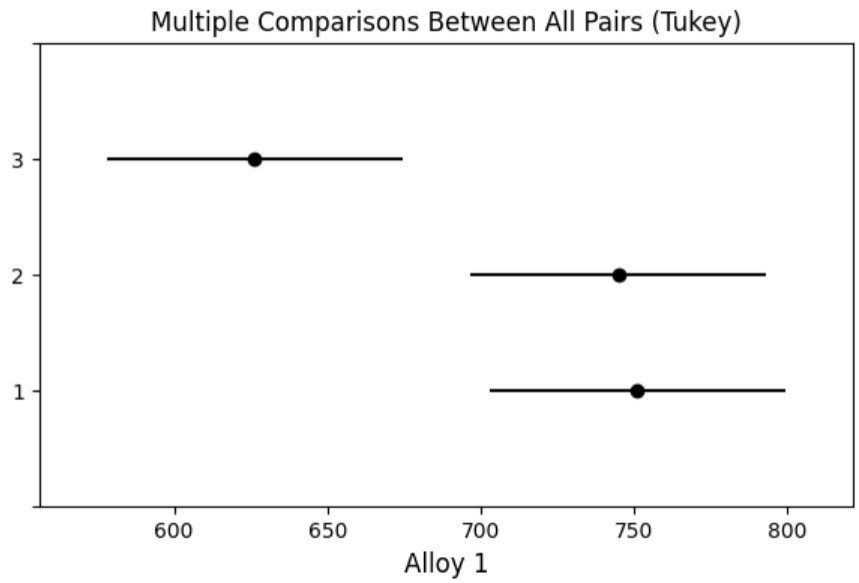
Conclusion:

Since the p-value is less than the level of significance (5%), we **Reject the null hypothesis**. Hence, we have enough statistical evidence to say that the mean hardness of implants is different for at least one Method.

Tukey HSD Test:

Alloy 1:

```
  Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2  meandiff p-adj    lower    upper   reject
-----------------------------------------------------
     1      2   -6.1333  0.987  -102.714  90.4473  False
     1      3    -124.8 0.0085 -221.3807 -28.2193   True
     2      3 -118.6667 0.0128 -215.2473  -22.086   True
-----------------------------------------------------
```

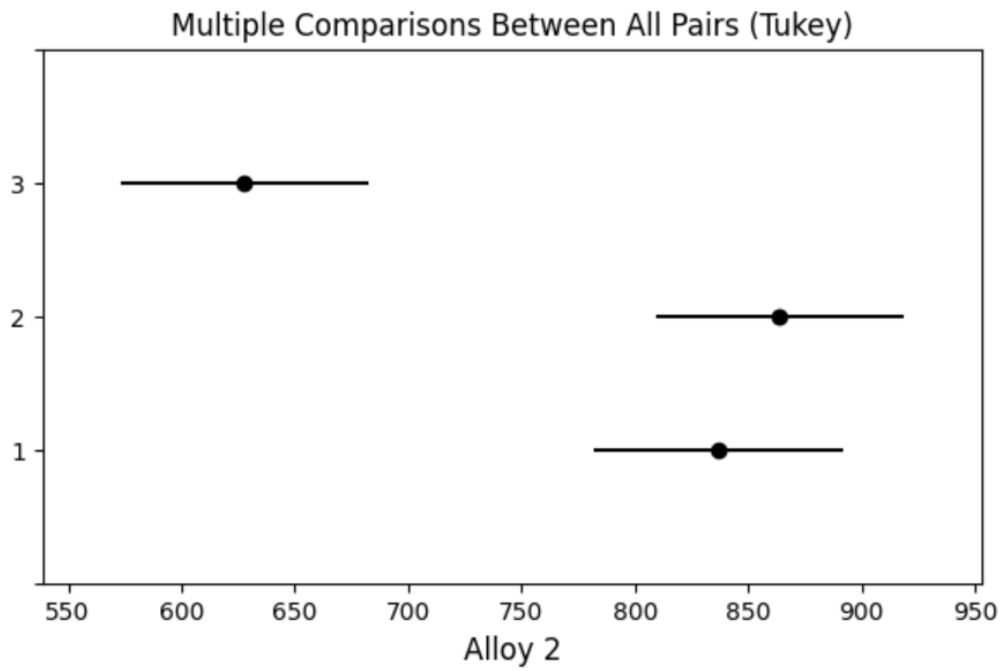


Inference:

Method 3 has significantly lower hardness compared to both Method 1 and Method 2. There is no significant difference between Method 1 and Method 2.

Alloy 2:

```
  Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj    lower     upper   reject
-----------------------------------------------------
     1      2     27.0 0.8212  -82.4546  136.4546  False
     1      3   -208.8 0.0001 -318.2546  -99.3454   True
     2      3   -235.8    0.0 -345.2546 -126.3454   True
-----------------------------------------------------
```

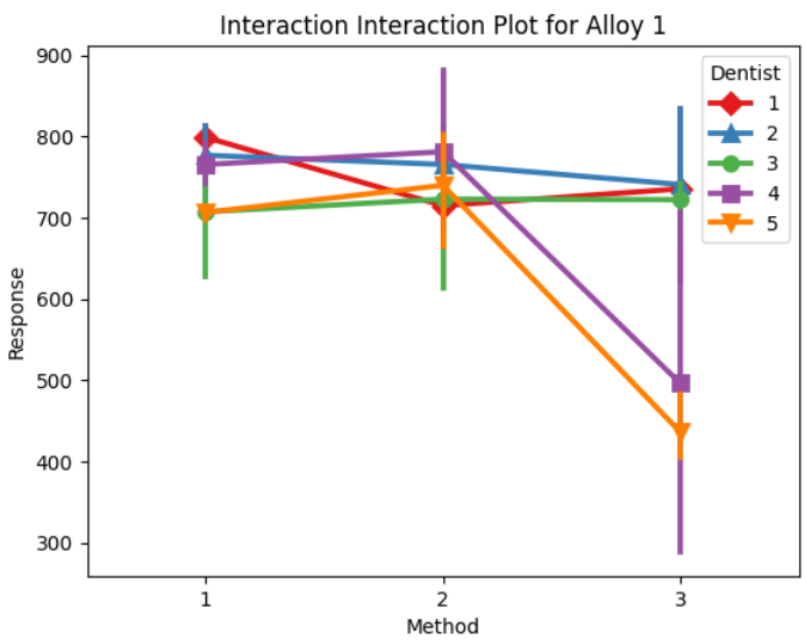


Inference:

Method 3 has significantly lower hardness compared to both Method 1 and Method 2. There is no significant difference between Method 1 and Method 2.

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

Alloy 1:

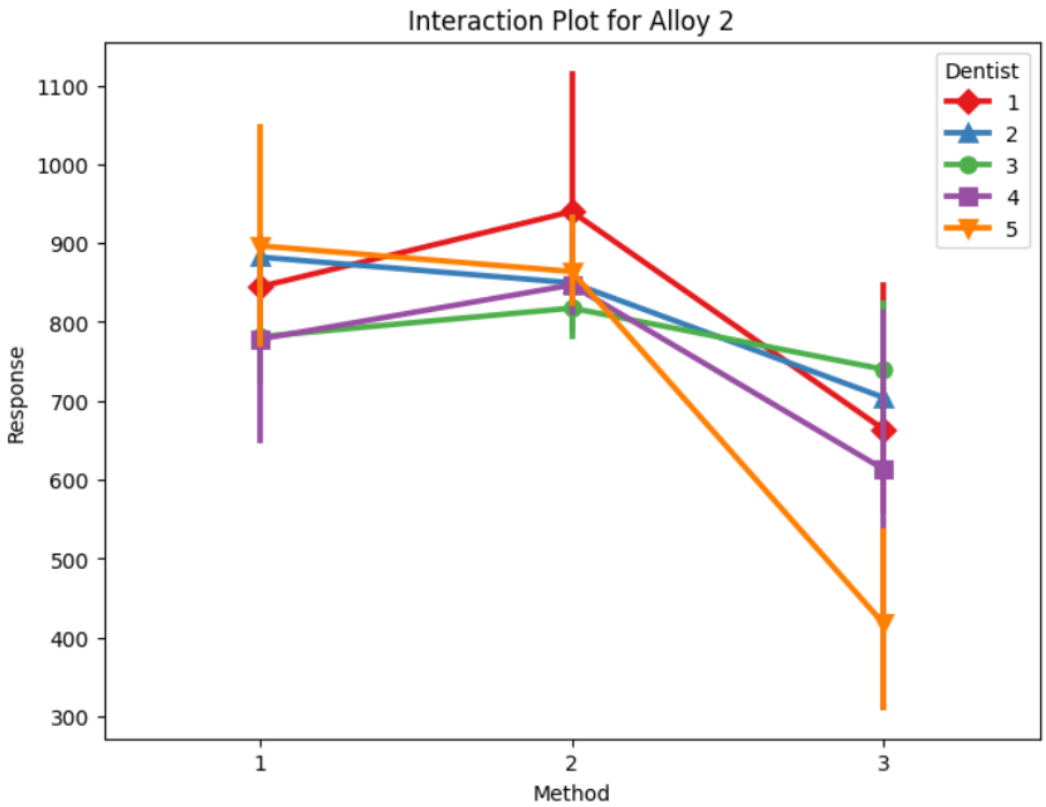


Inference:

- The Interaction plot for Alloy 1 shows how the hardness of implants varies with different combinations of dentists and methods. The lines connecting the points for each dentist indicate the interaction effect. If the lines are not parallel, it suggests a significant interaction effect between dentists and methods on the hardness of implants.

- If the lines cross, some dentists work better with certain methods; if they are parallel, the method affects all dentists the same way.

Alloy 2:



Inference:

- Similarly, the Interaction plot for Alloy 2 shows the interaction

effect between dentists and methods on the hardness of implants. Non-parallel lines indicate a significant interaction effect.

- If the lines move apart or cross, different dentists get different hardness results depending on the method used.

4.4 How does the hardness of implants vary depending on dentists and methods together?

Hypothesis:

- **Null Hypothesis (H0):** There is no interaction effect between Dentist and Method on the hardness of implants.
- **Alternative Hypothesis (H1):** There is a significant interaction effect between Dentist and Method on the hardness of implants.

Two -Way ANOVA Test:

Alloy 1:

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	106683.688889	26670.922222	3.899638	0.011484
C(Method)	2.0	148472.177778	74236.088889	10.854287	0.000284
C(Dentist):C(Method)	8.0	185941.377778	23242.672222	3.398383	0.006793
Residual	30.0	205180.000000	6839.333333	NaN	NaN

Test Results:

- **Dentist:** 0.011484 (p-value)
- **Method**: 0.000284 (p-value)
- **Interaction (Dentist * Method):** 0.006793 (p-value)

Inferences:

- As the p-value 0.006793 is less than the level of significance, we reject the null hypothesis.
- Both dentists and methods significantly affect the hardness of dental implants. There is also a significant interaction effect between dentists and methods.

Alloy 2:

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	56797.911111	14199.477778	1.106152	0.371833
C(Method)	2.0	499640.400000	249820.200000	19.461218	0.000004
C(Dentist):C(Method)	8.0	197459.822222	24682.477778	1.922787	0.093234
Residual	30.0	385104.666667	12836.822222	NaN	NaN

Test Results:

- **Dentist:** 0.371833 (p-value)
- **Method**: 0.000004 (p-value)
- **Interaction (Dentist * Method):** 0.093234 (p-value)

Inferences:

- As the p-value 0.09323403966149683 is greater than the level of significance, we fail to reject the null hypothesis.
- Methods significantly affect the hardness of dental implants. There is no significant effect of dentists or their interaction with methods.